CLONED_KMPh

PART A:

1.

a. 6

b. $\sqrt{\frac{5}{6}}$ or $\frac{\sqrt{30}}{6}$

c. $-\infty$

2.

a. $\frac{2x}{x^2-1}-\frac{1}{2(x-1)}$

b. -0.5931

c. $\frac{dy}{dx}=\frac{-6x^2-y}{x-6y}$

3.

a. $\text{Stationary points}=(-3,0),\ (1,32)$

b. $(-3,0)$ is a minimum point, $(1,32)$ is a maximum point

PART B

1.

a. $-\frac{1}{10}i$

b. $-\frac{\pi}{2}$

2.

a. $3-\frac{2}{3}\sqrt{3}$

b. $x=1, x=5.196$

c. $\{x:-3\le x\le 1\}$

3.

a. $a=\frac{1}{3}$ **b =-2 or**

$a=1$**, b =0**

b. i. ln 2 ii. $\text{Domain}=(2,\infty)$, $\text{Range}=(1,\infty)$

c. Domain = $\mathbb{R}$, Range = $[-3,3]$

d. $c = \dfrac{24}{7}$

4.

a. a = 8, b = 4

b. shown

5. $y = 1, y = -1$

6.

a. Shown , $\dfrac{20t^3}{\left(3t^2+1\right)^3}$

b. Shown, deduced

7. Width = $\dfrac{14}{3}$ length = $\dfrac{35}{3}$, height = $\dfrac{5}{3}$